

# Fructose: It's "Alcohol Without the Buzz"

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What do the Atkins Diet and the traditional Japanese diet have in common? The Atkins Diet is low in carbohydrate and usually high in fat; the Japanese diet is high in carbohydrate and usually low in fat. Yet both work to promote weight loss. One commonality of both diets is that they both eliminate the monosaccharide fructose. Sucrose (table sugar) and its synthetic sister high fructose corn syrup consist of 2 molecules, glucose and fructose. Glucose is the molecule that when polymerized forms starch, which has a high glycemic index, generates an insulin response, and is not particularly sweet. Fructose is found in fruit, does not generate an insulin response, and is very sweet. Fructose consumption has increased worldwide, paralleling the obesity and chronic metabolic disease pandemic. Sugar (i.e., fructose-containing mixtures) has been vilified by nutritionists for ages as a source of "empty calories," no different from any other empty calorie. However, fructose is unlike glucose. In the hypercaloric glycogen-replete state, intermediary metabolites from fructose metabolism overwhelm hepatic mitochondrial capacity, which promotes de novo lipogenesis and leads to hepatic insulin resistance, which drives chronic metabolic disease. Fructose also promotes reactive oxygen species formation, which leads to cellular dysfunction and aging, and promotes changes in the brain's reward system, which drives excessive consumption. Thus, fructose can exert detrimental health effects beyond its calories and in ways that mimic those of ethanol, its metabolic cousin. Indeed, the only distinction is that because fructose is not metabolized in the central nervous system, it does not exert the acute neuronal depression experienced by those imbibing ethanol. These metabolic and hedonic analogies argue that fructose should be thought of as "alcohol without the buzz."